8.4 Finding Sums of Infinite Geometric Series

Name _____ Period ____ Date _____

Algebra II 8.4 · Student sheet

I can…

- find partial sums of an infinite geometric series.
- find the sum of an infinite geometric series when $|r| < 1$.
- write a repeating decimal as a fraction.

Vocabulary · fill in the missing word

Partial sum S_n — the sum of the first n terms of an _____ series

Infinite geometric series — _____

Sum — S = _____ when $|r| < 1$

Diverges — an infinite _____ series with $|r| \geq 1$ has no sum

Part 1 · Walk to the wall

Start 8 feet from the wall. Each step goes halfway to the wall.

Step	1	2	3	4	5	6
Length (ft)						
Total walked (ft)						

If the walk never stopped, what would the total be? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Find the partial sums S_1 through S_5 of $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots$ What do the partial sums approach?

Sum of an infinite geometric series: When $|r|$ _____ 1, the sum is S = _____.

No sum: When $|r| \geq 1$, the series has _____.

Example 2

Find the sum of each infinite geometric series, if it exists.

a. Σ (i = 1 to ∞) $5(0.6)^{i-1}$ b. $1 + 3 + 9 + 27 + \ldots$ c. $6 - 2 + \frac{2}{3} - \frac{2}{9} + \ldots$

Repeating decimals: $0.444\ldots = 0.4 + 0.04 + \ldots$, a_1 = _____, r = _____

Example 3

Write each repeating decimal as a fraction in simplest form.

a. $0.444\ldots$ b. $0.363636\ldots$

Example 4

A pendulum swings 18 inches on its first swing. Each swing after that is 80% as long as the one before. What total distance does the pendulum swing?

Part 3 · Partner practice

Partner 1

Find the sum: $8 + 4 + 2 + 1 + \ldots$

Partner 2

Write $0.272727\ldots$ as a fraction in simplest form.